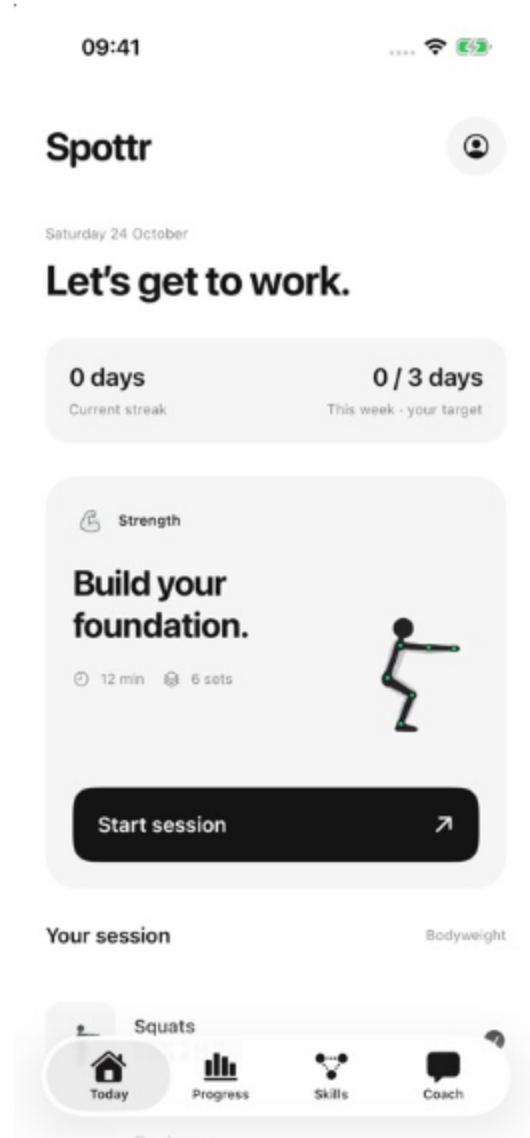
# Spottr

## Bodyweight coaching on iPhone and Watch

Short workout feedback needs fresh evidence.
The user stays in control of the set.

**OxHack 2026**



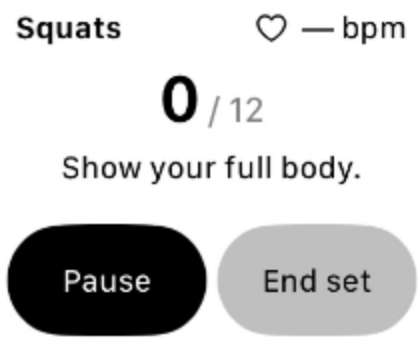

iPhone simulator

# The workout flow

| | | |
|---|---|---|
| 01 | **Choose the work** | Goals, weekly target and an approved workout. |
| 02 | **Track the set** | Camera cycles for push-ups and squats. Manual logs and hold timers for the wider library. |
| 03 | **Respond in the moment** | A concise cue, visible controls and optional local voice. Pause or stop at any time. |
| 04 | **Review the record** | Actual sets, source labels and timings. Ask for an AI summary with an explicit tap. |

# Phone, Watch and optional side camera




Simulator capture
Physical sensors pending

## Session authority

On-device pose inference
Rep and rest timing
Cue priority and expiry
Local workout history

## Capture companion

HTTPS pairing verified
Real initial frames
USB disconnect blocks sustained capture

# Timing decisions use observed evidence

### Complete cycles count

Top, bottom, then top. A gap above 0.75s resets movement.

### Pacing uses prior reps

After five valid reps, flag duration above mean + max(sample SD, 0.25s).

### Stop takes priority

Reported discomfort overrides ordinary cues. Pause and end cancel speech and old cues.

### Sensor time stays explicit

Missing or stale Watch readings remain unavailable.



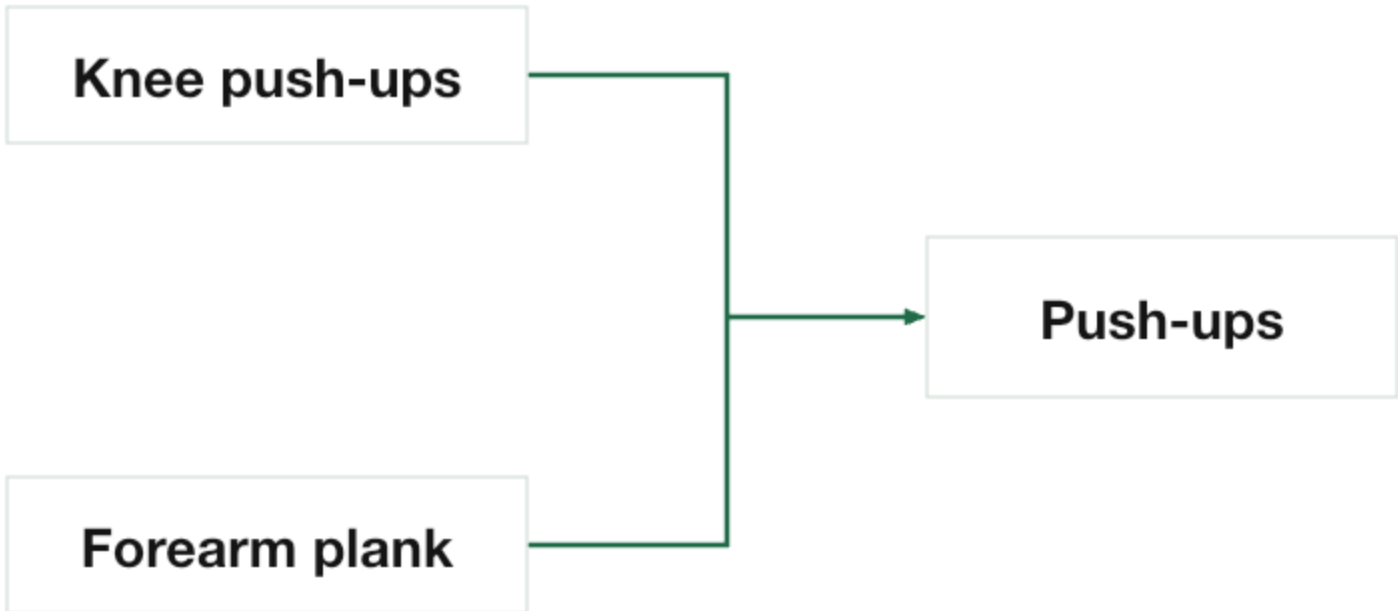




Illustrated phases

# Skills make recorded progress visible

19 exercises. Connected tree or linear list.




## 3 qualifying sets across 2 days

All prerequisites complete.
Practice remains open.
Sources stay visible.

Replay and discomfort
do not earn completion.

**One unlock example**

# Four Gemini requests share a grounded contract

01 **Chat**

02 **Workout proposal**

03 **Session summary**

04 **Weekly summary**

**Explicit facts and preferences**

Recorded sets and source labels
Selected goals and weekly target
Canonical catalogue capabilities

**Checked before use**

Fact IDs and typed quantities
Current generation only
User approval before a plan starts

4 gateway types; 2 native chats. Live native plans/reports and physical phone checks pending.

# Operational data and batch analysis

## Tiger Data

**Operational history**

Implemented adapters
Timestamped workout events
and rolling aggregates

Adapter and migration tests pass.
Strict TLS blocks live SQL.

## Snowflake

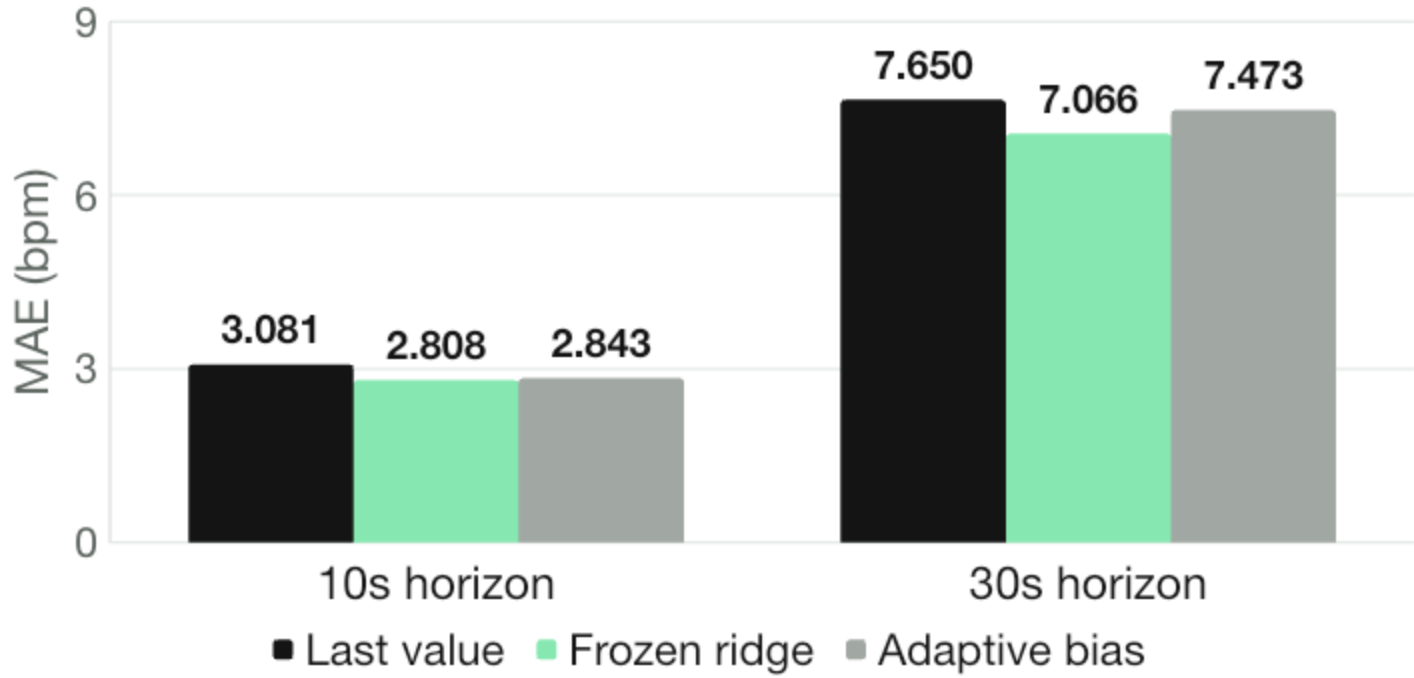
**Batch evaluation**

Implemented public-data batch
Pinned forecast lineage
and error reconciliation

Dry-run import/evaluation tested.
Live warehouse acceptance pending.

# An HR forecast with a locked participant test

Future 5s mean HR, using the previous 60 seconds




**Test coverage**

**Participants 108 / 109**

10s: 803 + 4 forecasts
30s: 799 + 0 forecasts

**The adapter increased error**

Exploratory public-data result. Only one person contributes at 30s. No Watch or failure validation.

# Provider evidence and remaining gates

| Track | Evidence now | Remaining gate |
| --- | --- | --- |
| Gemini | Gateway + 2 native chats | Native plans/reports; physical networking |
| Tiger Data | Service provisioned | Strict TLS private-CA chain |
| Snowflake | Pinned batch workflow tested | Existing warehouse and live SQL |
| Vultr | Local ARM64 runtime checked | Spend scope; hosted TLS/auth |
| ElevenLabs | Local Apple speech only | API secret and audible acceptance |
| Presage | SDK plan prepared | Vitals egress decision and credits |
| Delta | Judge packet prepared | Operator/legal step, then one review |

Standalone web prototype: 92 offline checks and mock-profile browser acceptance. Real cloud upload pending.

# A two-minute demo of the implemented flow

| | | |
|---|---|---|
| **00:00** | **Today and goals** | Show the actual starting state. |
| **00:15** | **A tracked workout** | Show the camera path and visible cue. Label any fixture. |
| **00:55** | **Pause, end and review** | Keep controls available and preserve source labels. |
| **01:20** | **Skills and Coach** | Show prerequisites, then an explicit request action. |
| **01:45** | **Evidence and next gates** | Use the saved HR result and current provider status. |

Builds 1 + 2: available internally in TestFlight.
External build 1: beta review pending; physical installs unverified.